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	(1) Original Aims: Our proposal contained two specific aims: (1) Determine whether elevations in endogenous Glucose-dependent insulinotropic peptide (GIP) levels lead to an increase in bone formation by examining the bone phenotype in GIP-overexpressing transgenic mice and (2) Determine whether endogenous elevations in GIP prevent bone loss in a model of weightlessness by measuring bone turnover rate in GIP-overexpressing transgenic mice with hindlimb suspension. (2) Key findings of the project Studies characterizing the GIP-overexpressing transgenic mice Aim 1 was broken down into four separate components in which we proposed to: (1) Characterize a transgenic mouse model overexpressing GIP (2) Evaluate the effects of GIP on bone loss associated with estrogen and androgen deficiency (3) Evaluate the effects of nutritional alterations on GIP		

Task Description:	effects on bone and (4) Evaluate the interaction between hormonal alterations and GIP effects on bone. Studies on specific aim 1 were to take place over the first two years of funding. Our initial strategy was to develop transgenic mice overexpressing GIP, to examine the bone phenotype in the transgenic lines. In view of data that the GIP receptor can be downregulated by high doses of GIP, it was necessary to design a construct that contained a regulatable promoter. Thus, the regulatory elements associated with the mouse metallothionein promoter which have been characterized extensively by Palmiter and colleagues (4, 5), were used. This promoter was induced by addition of 25mM zinc (ZnSO4) to the drinking water. In fact, addition of zinc resulted in a close to six-fold rise in GIP serum levels compared to that of control animals. While we had hoped that GIP levels in these transgenic mice would rise only in response to feeding with a heavy metal (zinc) we found that the transgenic animals had more than twice as much circulating GIP as nontransgenic controls basally and rose almost six-fold in response to zinc feeding consistent with a “leaky” promoter. Thus, the transgenic animals had a significantly higher bone density even before zinc feeding. Interestingly, biomechanically, these mice also had stronger bones. We expected our GIP-overexpressing transgenic animals to be protected from bone loss when placed on a low calcium diet. Twenty four mice were divided into four equal groups: (1) non-transgenic littermate controls (2) non transgenic littermate controls placed on a low calcium diet (3) transgenic mice GIP overexpressing mice (4) transgenic mice overexpressing GIP on a low calcium diet. Weanling mice were fed a standard semipurified diet based on the American Institute of Nutrition recommendations. Drinking water was supplemented with 25mM ZnSO4 for 40 weeks. A number of gross measurements were obtained bi-weekly during the five months included: body weight, body length, serum GIP levels and bone density. Additionally, markers of bone formation and breakdown were examined. Samples for serum calcium, parathyroid hormone, insulin, osteocalcin and pyridinium cross-links were collected from each mouse at the beginning and end of the study period. Serum for GIP analysis was obtained by retroorbital bleeding and bone density was measured by DEXA. At the end of the two-month experimental protocol animals were sacrificed and three animals of each group had their vertebrae, tibiae and femurs removed and prepared for bone histomorphometrical analysis). The remaining skeletons were processed and stored. Paradoxically, the GIP overexpressing transgenic mice were found to be more prone to lose bone after three months of a low calcium diet. The data would suggest that in the case of GIP, as is the case for other peptide hormone receptors, continuous elevations leads to GIP receptor downregulation. Furthermore if this were the case, then we would predict that mice in which the GIP receptor was downregulated, there would be an increased susceptibility to bone loss. To evaluate these possibilities we did a Western blot for GIP receptor and found that in fact that after five months of high GIP levels the GIP-overexpressing transgenic mice had GIPR downregulation . Furthermore, if these mice were placed on a low calcium diet they were more prone to lose bone than control mice. Taken together these data suggest an important role for GIP in bone formation since supraphysiological elevations of GIP lead to increased bone mass and pharmacological concentrations of GIP lead to GIP receptor downregulation and bone loss. (2) Aim 2 was to Determine whether endogenous elevations in GIP prevent bone loss in a model of weightlessness by measuring bone turnover rate in GIP-overexpressing transgenic mice with hindlimb suspension. The data for these studies has not yet been fully analyzed, however at 3 months of age, GIP overexpressing transgenic mice do not appear to be protected against bone loss when tail suspendes (as a model of microgravity). However, in a joint project with another NSBRI project (Dr. Joseph Zerweck) where we examined changes in GIP levels during bedrest we found that bed rest altered the dynamics of GIP secretion. In addition, subsequently we have found that with time as the mice age, the GIP effects on bone mass continue to increase thus, the initial selection of 3 month old mice for tail suspension was probably not the optimal age for these studies. These studies will be repeated with 12 month old mice which is an age closer to peak bone mass for this strain. (3) The impact of these findings on the hypotheses objectives and specific aims of the original proposal The findings of the past year validate our hypothesis that GIP could be a useful countermeasure to microgravity-induced bone loss. Our data is consistent with GIP being the hormone linking nutrient intake and bone formation. (4) The proposed research plan for the coming year This project has completed its three years of funding.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a disease characterized by an imbalance between bone formation and breakdown leading to a decrease in bone mass, resulting in an increased risk of fractures. Osteoporosis is associated with significant morbidity and mortality due to fractures, in particular vertebral compression fractures and hip fractures. A number of epidemiological and clinical studies have linked abnormalities in nutrition to osteoporosis. Thus, although there is a considerable amount of clinical data to link abnormalities in nutritional intake with osteoporosis, very little research has delved into this phenomenon. Present therapy for osteoporosis is predominantly antiresorptive therapy, i.e. to prevent further bone breakdown and few medications (except for parathyroid hormone injection, perhaps estrogen in women and fluoride) are able to increase new bone formation. Our studies should shed further light on the mechanisms involved in the development of osteoporosis, and thus help in the development of new anabolic therapies. Our data demonstrates that GIP is a key hormone linking nutrient intake to bone formation. If GIP levels are elevated then increases in bone mass are observed. However if the elevation in GIP level is prolonged or excessive then this results in loss of the GIP effect and loss of bone mass.
Task Progress:	Major new findings during this past year of funding include: (1) Validation of our in vitro findings suggesting a role for GIP as a link between nutrient intake and bone formation. We now have two animal models: (a) a GIP-overexpressing transgenic mouse which has a higher bone mass which, the indices demonstrate, have higher bone formation and decreased bone breakdown and (b) a functional GIP receptor knockout mouse, generated by chronically exposing them to high GIP levels. These mice have a lower bone mass, higher indices of bone breakdown and lower indices of bone formation. These mice are more prone to bone loss. These data suggest that GIP levels may be an important risk factor for development of osteoporosis and validates our original proposal that GIP might be a valuable countermeasure to microgravity-induced bone loss. (2) Demonstration of GIP as a major nutritional “integrative” hormone capable to affect multiple tissues to maximize

	nutrient utilization. Thus, manipulation of GIP for bone maintenance in astronauts in space may have the additional benefit of increasing nutrient utilization and maximize nutritional health. (3) Demonstration that GIP interactions with other anabolic hormones such as PTH on bone mass are complex with some indices (ex. alkaline phosphatase) rising more with the combination but not other parameters (ex. bone densitometry).
Bibliography Type:	Description: (Last Updated:)
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